

Vinay Kumar Mandadi

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EDUCATION

Lovely Professional University (LPU), Punjab, India Expected 2026
B.Tech in Computer Science and Engineering
Focus Areas: Cloud Computing, Distributed Systems, DevOps, Applied Machine Learning
Relevant Coursework: Cloud Computing, Distributed Systems, Machine Learning, Site Reliability Engineering

RESEARCH INTERESTS

AI/ops, Cloud Reliability Engineering, Predictive Machine Learning for Distributed Systems, Explainable AI, Infrastructure-as-Code

RESEARCH WORK

Combating Synthetic Media Manipulation Using Deep Learning-Based Deepfake Detection Systems Accepted at 8BIC 2026

Co-author. Developed a deep learning framework (EfficientNetV2-M + CNN on DCT frequency) achieving 94.4% accuracy on FaceForensics++ for synthetic media authentication.

[\[Project Repository\]](#)

AIRIMF: AI-Driven Risk Identification and Mitigation Framework

Sole author

Leakage-free Random Forest pipeline with SMOTE for imbalanced software risk prediction using real-world GitHub ticket metadata (title, body, comments, NLP sentiment). Achieved 76% cross-validation accuracy on VS Code, 83% on React.

[\[Project Repository\]](#)

NETOPS-XAI: Explainable Multi-Class Intrusion Detection

Sole author

SHAP-based intrusion detection framework on CIC-IDS2017 (98.8% in-domain accuracy). Quantified catastrophic domain shift to UNSW-NB15 (31.9% accuracy, 0% attack recall). Applied Platt scaling to boost out-of-domain accuracy to 68.1%, but confidence-based filtering still fails.

[\[Project Repository\]](#)

PROFESSIONAL CERTIFICATIONS

- Oracle Cloud Infrastructure (OCI) 2025 – Certified Multicloud Architect Professional
- Oracle Cloud Infrastructure (OCI) 2025 – Certified DevOps Professional
- Oracle Cloud Infrastructure (OCI) 2025 – Certified AI Foundations Associate
- Oracle Cloud Infrastructure (OCI) 2025 – Certified Data Platform Foundations
- Google Cloud Data Analytics Certificate
- AWS Cloud Practitioner Essentials

TECHNICAL SKILLS

Cloud & DevOps OCI, AWS, GCP, Terraform, Docker, Kubernetes, CI/CD, Firebase, Infrastructure-as-Code
Machine Learning & AI Python, scikit-learn, pandas, NumPy, Random Forest, SMOTE, SHAP, Feature Engineering, PyTorch, TensorFlow

Programming Languages

Python, Java, Bash, TypeScript, JavaScript

Monitoring & Tools

Prometheus, Grafana, Git, Linux, Nginx

TEST SCORES & LANGUAGE PROFICIENCY

GRE General Test

324 (Quantitative: 168, Verbal: 156)

English

Full professional proficiency (MOI certificate available)

Japanese

Elementary proficiency; self-studying toward JLPT N4

Native Languages

Telugu, Hindi

TECHNICAL PROJECTS

Carpooling and RideShare Platform

Full-stack ride-sharing web app using Node.js, Express.js, MongoDB, session-based authentication with ride creation, booking, and review modules.

StockAI

Stock market analysis web app using TypeScript and Firebase hosting with interactive frontend dashboards and market visualisation.

Secure Aadhaar Voting System

Cloud-based voting platform using Google Cloud (Cloud Storage, Cloud SQL, IAM) with Aadhaar-authenticated voting workflow.

Compass AI Travel Web Application

AI-assisted travel planning web app using Firebase hosting, real-time sync, and personalised itinerary generation.

TRAINING & COURSEWORK

Google Cloud ACE Training Outstanding grade – focused on Terraform, Infrastructure-as-Code, and Cloud Operations.

AWS Training Completed AWS fundamentals and hands-on labs via Programming Pathasala.

ACHIEVEMENTS

- GDG Solution Challenge – National Finalist (Top 3% of 3,700+ teams)
- Google Cloud Skills Boost – Gold Tier
- DSA Problem Solving – 200+ problems on LeetCode & GeeksForGeeks